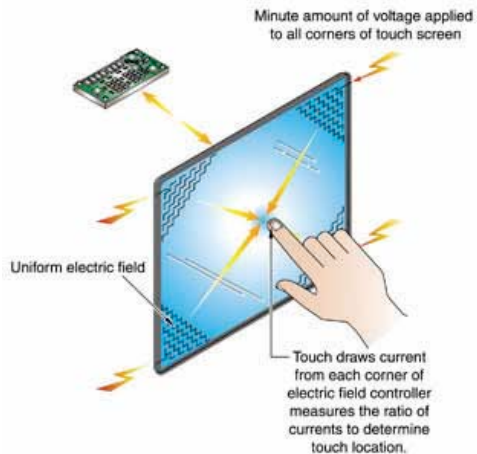


into an island of a device. They were priced very high, around \$800 which was a very steep price for its feature set, particularly at the time. Lastly, after Steve Jobs returned to Apple, he showed no interest in continuing development on the Newton platform, and development officially ended in 1993.

Methods of data entry

The tablets and mobile devices such as PDAs and smartphones use many different technologies to implement their core function – data entry. While some use a resistive touch screen along with a stylus for handwriting or virtual keyboards, others choose to use purely capacitive touch screens that require the use of one's fingers only. Still others come with full physical keyboards for quick data entry.

Early tablets such as the PenPoint display and the Apple Newton used a 'resistive touchscreen' which consisted of two thin conducting sheets separated by a distance. When a point on the screen was pressed down, the pressure would cause these two thin sheets to deform and come in contact, thereby completing an electric circuit. This would then act as a voltage divider, and the voltage sensed at the other end of the screen would help in determining the exact location of the touch. Since the touch relies purely on pressure, the technology is device agnostic, i.e. It doesn't matter whether you use a stylus or your fingers to interact with it. Styluses were popular due to their ability to precisely pinpoint a location as well as aid in handwriting. However, the styluses were an auxiliary component that had to be carried around with the phone everywhere, and were liable to be misplaced or forgotten, in which case the user would be in trouble. The technology of resistive touchscreens itself had many inherent flaws. A significant portion of the light emitted by the display was absorbed by the screen itself, resulting in lower brightness of the display. Since the display only registered a change in pressure on the screen,



The principle behind the working of a capacitive touchscreen